Set A:

1. Create a custom "Contact" layout to hold multiple pieces of information, including: Photo, Name, Contact Number, E-mail id and display it using fragment.

Fragment Layout (`fragment_contact.xml`)

This is the **custom contact UI**.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Contact Photo -->
    <ImageView
        android:id="@+id/imgPhoto"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@drawable/ic_person"
        android:layout_gravity="center"
        android:contentDescription="Contact Photo"/>

    <!-- Name -->
    <TextView
        android:id="@+id/txtName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginTop="12dp"/>

    <!-- Phone -->
    <TextView
        android:id="@+id/txtPhone"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Contact Number"
        android:layout_marginTop="8dp"/>

    <!-- Email -->
    <TextView
        android:id="@+id/txtEmail"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Email ID"
        android:layout_marginTop="8dp"/>

</LinearLayout>
```

📌 Add a placeholder image like `ic_person.png` in `res/drawable`

2. Contact Fragment (`ContactFragment.java`)

```
package com.example.contactapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class ContactFragment extends Fragment {

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
                             @Nullable Bundle savedInstanceState) {

 View view = inflater.inflate(R.layout.fragment_contact, container, false);

        ImageView imgPhoto = view.findViewById(R.id.imgPhoto);
        TextView txtName = view.findViewById(R.id.txtName);
        TextView txtPhone = view.findViewById(R.id.txtPhone);
        TextView txtEmail = view.findViewById(R.id.txtEmail);

        // Setting contact details
        imgPhoto.setImageResource(R.drawable.ic_person);
        txtName.setText("Amruta More");
        txtPhone.setText("Phone: 9876543210");
        txtEmail.setText("Email: amruta.more@gmail.com");

        return view;
    }
}
```

3. Activity Layout (`activity_main.xml`)

This holds the fragment.

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

4. Main Activity (`MainActivity.java`)

Loads the fragment.

```
package com.example.contactapp;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ContactFragment contactFragment = new ContactFragment();

        FragmentTransaction transaction =
getSupportFragmentManager().beginTransaction();
        transaction.replace(R.id.fragment_container, contactFragment);
        transaction.commit();
    }
}
```

---

2) Create an application to add and remove items from spinner. Use Spinner, Text box and Buttons to draw the GUI.

1. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Text Box -->
    <EditText
        android:id="@+id/edtItem"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter item"/>

    <!-- Spinner -->
    <Spinner
        android:id="@+id/spinnerItems"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"/>

    <!-- Add Button -->
    <Button
        android:id="@+id/btnAdd"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Add Item"
        android:layout_marginTop="16dp"/>

    <!-- Remove Button -->
    <Button
        android:id="@+id/btnRemove"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Remove Item"
        android:layout_marginTop="8dp"/>

</LinearLayout>
```

---

2. Main Activity (`MainActivity.java`)

```
package com.example.spinnerapp;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    Spinner spinner;
    EditText edtItem;
    Button btnAdd, btnRemove;

    ArrayList<String> itemList;
    ArrayAdapter<String> adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        spinner = findViewById(R.id.spinnerItems);
        edtItem = findViewById(R.id.edtItem);
        btnAdd = findViewById(R.id.btnAdd);
        btnRemove = findViewById(R.id.btnRemove);

        // Initial data
        itemList = new ArrayList<>();
        itemList.add("Apple");
        itemList.add("Banana");
        itemList.add("Orange");

        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_item,
                itemList
        );

adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_it
em);
        spinner.setAdapter(adapter);

        // Add item
        btnAdd.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
            public void onClick(View view) {
                String item = edtItem.getText().toString().trim();

                if (!item.isEmpty()) {
                    itemList.add(item);
                    adapter.notifyDataSetChanged();
                    edtItem.setText("");
                } else {
                    Toast.makeText(MainActivity.this,
                            "Enter an item", Toast.LENGTH_SHORT).show();
                }
            }
        });

        // Remove selected item
        btnRemove.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                if (itemList.size() > 0) {
                    int position = spinner.getSelectedItemPosition();
                    itemList.remove(position);
                    adapter.notifyDataSetChanged();
                } else {
                    Toast.makeText(MainActivity.this,
                            "No items to remove",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

---

3. Create application to demonstrate date and time picker. Use datePickerDialog and timePickerDialog

1. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Date Text -->
    <TextView
        android:id="@+id/txtDate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Selected Date"
        android:textSize="18sp" />

    <!-- Pick Date Button -->
    <Button
        android:id="@+id/btnDate"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:text="Select Date"
        android:layout_marginTop="12dp"/>

    <!-- Time Text -->
    <TextView
        android:id="@+id/txtTime"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Selected Time"
        android:textSize="18sp"
        android:layout_marginTop="24dp"/>

    <!-- Pick Time Button -->
    <Button
        android:id="@+id/btnTime"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Time"
        android:layout_marginTop="12dp"/>

</LinearLayout>
```

---

2. Main Activity (`MainActivity.java`)

```
package com.example.datetimepicker;

import android.app.DatePickerDialog;
import android.app.TimePickerDialog;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    TextView txtDate, txtTime;
    Button btnDate, btnTime;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        txtDate = findViewById(R.id.txtDate);
        txtTime = findViewById(R.id.txtTime);
        btnDate = findViewById(R.id.btnDate);
        btnTime = findViewById(R.id.btnTime);

        Calendar calendar = Calendar.getInstance();

        // Date Picker
        btnDate.setOnClickListener(view -> {
            int year = calendar.get(Calendar.YEAR);
            int month = calendar.get(Calendar.MONTH);
            int day = calendar.get(Calendar.DAY_OF_MONTH);
```

```
            DatePickerDialog datePickerDialog = new DatePickerDialog(
                    MainActivity.this,
                    (datePicker, y, m, d) -> {
                   txtDate.setText("Date: " + d + "/" + (m + 1) + "/" + y);
                    },
                    year, month, day
            );
            datePickerDialog.show();
        });

        // Time Picker
        btnTime.setOnClickListener(view -> {
            int hour = calendar.get(Calendar.HOUR_OF_DAY);
            int minute = calendar.get(Calendar.MINUTE);

            TimePickerDialog timePickerDialog = new TimePickerDialog(
                    MainActivity.this,
                    (timePicker, h, m) -> {
                        txtTime.setText("Time: " + h + ":" + m);
                    },
                    hour, minute, true
            );
            timePickerDialog.show();
        });
    }
}
```

---

4. Construct an app that toggles a light bulb on and off when the user clicks on toggle button.

To create image: Goto res-> drawable

Right click and select new->Drawable resource files

Create file with .xml extension.

Steps:

1. Go to **res → drawable**
2. **Right click → New → Drawable Resource File**
3. Create two files:
    - `bulb_on.xml`
    - `bulb_off.xml`

`bulb_on.xml` (Light ON)

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">

    <solid android:color="#FFFF00" />
    <size
        android:width="200dp"
        android:height="200dp" />
</shape>
```

`bulb_off.xml` (Light OFF)

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">

    <solid android:color="#AAAAAA" />
    <size
        android:width="200dp"
        android:height="200dp" />
</shape>
```

---

2. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <!-- Light Bulb Image -->
    <ImageView
        android:id="@+id/imgBulb"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/bulb_off" />

    <!-- Toggle Button -->
    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Light ON"
        android:textOff="Light OFF"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

---

3. Main Activity (`MainActivity.java`)

```
package com.example.lighttoggleapp;

import android.os.Bundle;
import android.widget.ImageView;
import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ImageView imgBulb;
    ToggleButton toggleButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        imgBulb = findViewById(R.id.imgBulb);
        toggleButton = findViewById(R.id.toggleButton);

        toggleButton.setOnCheckedChangeListener((buttonView, isChecked) ->
{
            if (isChecked) {
                imgBulb.setImageResource(R.drawable.bulb_on);
            } else {
                imgBulb.setImageResource(R.drawable.bulb_off);
            }
        });
    }
}
```

---

5. Create an Android application that displays an image based on the current date.

1. Add Images to Drawable Folder

Place 3 images in **res → drawable**:

- `img1.png`
- `img2.png`
- `img3.png`

(You can use any images.)

---

2. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:id="@+id/txtDate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Today Date"
        android:textSize="18sp"
        android:layout_marginBottom="20dp"/>

    <ImageView
        android:id="@+id/imgDate"
        android:layout_width="250dp"
        android:layout_height="250dp"
        android:contentDescription="Date Image"/>
</LinearLayout>
```

3. Main Activity (`MainActivity.java`)

```
package com.example.dateimageapp;

import android.os.Bundle;
import android.widget.ImageView;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    TextView txtDate;
    ImageView imgDate;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        txtDate = findViewById(R.id.txtDate);
        imgDate = findViewById(R.id.imgDate);

        // Get current date
        Calendar calendar = Calendar.getInstance();
        int day = calendar.get(Calendar.DAY_OF_MONTH);
        int month = calendar.get(Calendar.MONTH) + 1;
        int year = calendar.get(Calendar.YEAR);

        txtDate.setText("Date: " + day + "/" + month + "/" + year);

        // Display image based on date
        if (day >= 1 && day <= 10) {
            imgDate.setImageResource(R.drawable.img1);
        } else if (day >= 11 && day <= 20) {
            imgDate.setImageResource(R.drawable.img2);
        } else {
            imgDate.setImageResource(R.drawable.img3);
        }
    }
}
```

---

6. Create a custom launcher icon.

Steps to Create a Custom Launcher Icon

1. Open Your Android Project

Open your project in **Android Studio**.

---

2. Open Image Asset Tool

1. Go to **app → res**
2. **Right-click on `res`**
3. Select
   New → Image Asset

---

3. Configure Launcher Icon

In the **Configure Image Asset** window:

- **Icon Type**: `Launcher Icons (Adaptive and Legacy)`
- **Foreground Layer**:
  - Choose **Image** or **Clip Art**
  - Select your custom image (PNG/JPG)
- **Background Layer**:
  - Choose a color or image
- Adjust size and preview if needed

Click **Next → Finish**

📌 Android Studio automatically creates icons in:

- `mipmap-hdpi`
- `mipmap-mdpi`
- `mipmap-xhdpi`
- `mipmap-xxhdpi`
- `mipmap-xxxhdpi`

---

4. Verify AndroidManifest.xml

Make sure your app icon is set correctly:

```
<application
    android:icon="@mipmap/ic_launcher"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:label="@string/app_name"
    android:theme="@style/Theme.MyApp">
```

👉 Android Studio sets this automatically in most cases.

---

5. Run the App

- Install the app on emulator or device

- The **custom launcher icon** will appear on the home screen

Set B:

1. Create application to demonstrate file explorer (Use ListView).

Add permission in **AndroidManifest.xml** to read storage
(for Android 6–10; for newer versions this is for demo/academic use).

```
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>
```

---

1. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView
        android:id="@+id/txtPath"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="10dp"
        android:text="Current Path"
        android:textStyle="bold"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

---

2. Main Activity (`MainActivity.java`)

```
package com.example.fileexplorer;

import android.os.Bundle;
import android.os.Environment;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

import java.io.File;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    ListView listView;
```

```
    TextView txtPath;

    File currentDir;
    ArrayList<String> fileList;
    ArrayAdapter<String> adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);
        txtPath = findViewById(R.id.txtPath);

        // Start from external storage root
        currentDir = Environment.getExternalStorageDirectory();
        loadFiles(currentDir);

        listView.setOnItemClickListener(new
AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int
position, long id) {
                File selectedFile = new File(currentDir,
fileList.get(position));

                if (selectedFile.isDirectory()) {
                    currentDir = selectedFile;
                    loadFiles(currentDir);
                }
            }
        });
    }

    private void loadFiles(File dir) {
        txtPath.setText("Path: " + dir.getAbsolutePath());

        fileList = new ArrayList<>();
        File[] files = dir.listFiles();

        if (files != null) {
            for (File file : files) {
                fileList.add(file.getName());
            }
        }

        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                fileList
        );
        listView.setAdapter(adapter);
    }
}
```

---

2. Create registration form given. Also perform appropriate validation and display the message using dialog fragment.

1. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <EditText
            android:id="@+id/edtRoll"
            android:hint="Roll Number"
            android:inputType="number"/>

        <EditText
            android:id="@+id/edtName"
            android:hint="Name"/>

        <EditText
            android:id="@+id/edtDob"
            android:hint="Date of Birth"/>

        <EditText
            android:id="@+id/edtAddress"
            android:hint="Address"/>

        <EditText
            android:id="@+id/edtContact"
            android:hint="Contact Number"
            android:inputType="phone"/>

        <RadioGroup
            android:id="@+id/radioGender">
            <RadioButton android:text="Male"/>
            <RadioButton android:text="Female"/>
        </RadioGroup>

        <EditText
            android:id="@+id/edtEmail"
            android:hint="Email ID"
            android:inputType="textEmailAddress"/>

        <EditText
            android:id="@+id/edtPassword"
            android:hint="Password"
            android:inputType="textPassword"/>

        <Button
            android:id="@+id/btnRegister"
            android:text="Register"/>

    </LinearLayout>
</ScrollView>
```

---

2. Dialog Fragment (`MessageDialog.java`)

```
package com.example.registrationapp;

import android.app.AlertDialog;
import android.app.Dialog;
import android.os.Bundle;

import androidx.annotation.NonNull;
import androidx.fragment.app.DialogFragment;

public class MessageDialog extends DialogFragment {

    String message;

    public MessageDialog(String message) {
        this.message = message;
    }

    @NonNull
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        return new AlertDialog.Builder(getActivity())
                .setTitle("Message")
                .setMessage(message)
                .setPositiveButton("OK", null)
                .create();
    }
}
```

---

3. Main Activity (`MainActivity.java`)

```
package com.example.registrationapp;

import android.os.Bundle;
import android.util.Patterns;
import android.widget.*;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText edtRoll, edtName, edtDob, edtAddress, edtContact, edtEmail,
edtPassword;
    RadioGroup radioGender;
    Button btnRegister;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        edtRoll = findViewById(R.id.edtRoll);
        edtName = findViewById(R.id.edtName);
        edtDob = findViewById(R.id.edtDob);
        edtAddress = findViewById(R.id.edtAddress);
        edtContact = findViewById(R.id.edtContact);
        edtEmail = findViewById(R.id.edtEmail);
```

```
        edtPassword = findViewById(R.id.edtPassword);
        radioGender = findViewById(R.id.radioGender);
        btnRegister = findViewById(R.id.btnRegister);

        btnRegister.setOnClickListener(v -> validateForm());
    }

    private void validateForm() {

        if (edtRoll.getText().toString().isEmpty() ||
                edtName.getText().toString().isEmpty() ||
                edtDob.getText().toString().isEmpty() ||
                edtAddress.getText().toString().isEmpty() ||
                edtContact.getText().toString().isEmpty() ||
                edtEmail.getText().toString().isEmpty() ||
                edtPassword.getText().toString().isEmpty()) {

            showDialog("All fields are required");
            return;
        }

        if (edtContact.getText().toString().length() != 10) {
            showDialog("Invalid contact number");
            return;
        }
        if (!Patterns.EMAIL_ADDRESS.matcher(
                edtEmail.getText().toString()).matches()) {
            showDialog("Invalid email ID");
            return;
        }
        if (radioGender.getCheckedRadioButtonId() == -1) {
            showDialog("Please select gender");
            return;
        }
        showDialog("Registration Successful!");
    }
    private void showDialog(String message) {
        MessageDialog dialog = new MessageDialog(message);
        dialog.show(getSupportFragmentManager(), "msgDialog");
    }
}
```

---

3. Construct image switcher using SETFactory().

1. Add Images to Drawable Folder

Add a few images in **res → drawable**, for example:

- `img1.png`
- `img2.png`
- `img3.png`

---

2. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ImageSwitcher
        android:id="@+id/imageSwitcher"
        android:layout_width="300dp"
        android:layout_height="300dp"/>

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="20dp">

        <Button
            android:id="@+id/btnPrev"
            android:text="Previous"/>

        <Button
            android:id="@+id/btnNext"
            android:text="Next"
            android:layout_marginStart="20dp"/>
    </LinearLayout>

</LinearLayout>
```

---

3. Main Activity (`MainActivity.java`)

```
package com.example.imageswitcherapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageSwitcher;
import android.widget.ImageView;
import android.widget.ViewSwitcher;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ImageSwitcher imageSwitcher;
    Button btnPrev, btnNext;

    int[] images = {
            R.drawable.img1,
            R.drawable.img2,
            R.drawable.img3
    };

    int position = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        imageSwitcher = findViewById(R.id.imageSwitcher);
        btnPrev = findViewById(R.id.btnPrev);
        btnNext = findViewById(R.id.btnNext);

        // Set factory for ImageSwitcher
        imageSwitcher.setFactory(new ViewSwitcher.ViewFactory() {
            @Override
            public View makeView() {
                ImageView imageView = new ImageView(MainActivity.this);
                imageView.setScaleType(ImageView.ScaleType.CENTER_CROP);
                imageView.setLayoutParams(
                        new ImageSwitcher.LayoutParams(
                                ImageSwitcher.LayoutParams.MATCH_PARENT,
                                ImageSwitcher.LayoutParams.MATCH_PARENT));
                return imageView;
            }
        });

        // Set first image
        imageSwitcher.setImageResource(images[position]);

        btnNext.setOnClickListener(v -> {
            position = (position + 1) % images.length;
            imageSwitcher.setImageResource(images[position]);
        });

        btnPrev.setOnClickListener(v -> {
            position = (position - 1 + images.length) % images.length;
            imageSwitcher.setImageResource(images[position]);
        });
    }
}
```

---

4. Construct a bank app to display different menu like windrow, deposite etc.

1. Layout File (`activity_main.xml`)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Bank Menu"
        android:textSize="22sp"
```

```
        android:textStyle="bold"
        android:gravity="center"
        android:padding="10dp"/>

    <ListView
        android:id="@+id/listMenu"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

---

2. Main Activity (`MainActivity.java`)

```
package com.example.bankapp;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listMenu;
    String[] menuItems = {
            "Withdraw",
            "Deposit",
            "Balance Enquiry",
            "Mini Statement",
            "Exit"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listMenu = findViewById(R.id.listMenu);

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                menuItems
        );

        listMenu.setAdapter(adapter);

        listMenu.setOnItemClickListener((parent, view, position, id) -> {
            String selectedItem = menuItems[position];

            switch (selectedItem) {
                case "Withdraw":
                    Toast.makeText(this,
                            "Withdraw selected",
Toast.LENGTH_SHORT).show();
                    break;

                case "Deposit":
```

```
                    Toast.makeText(this,
                            "Deposit selected", Toast.LENGTH_SHORT).show();
                    break;

                case "Balance Enquiry":
                    Toast.makeText(this,
                            "Your balance is ₹10,000",
Toast.LENGTH_SHORT).show();
                    break;

                case "Mini Statement":
                    Toast.makeText(this,
                            "Mini Statement opened",
Toast.LENGTH_SHORT).show();
                    break;

                case "Exit":
                    finish();
                    break;
            }
        });
    }
}
```